James Yihan Zhang

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Education

Yale University — Sep 2025 – Present
Visiting Graduate Student: Financial Economics I, Econometrics I (First-Year PhD Courses) — *Grades: Honors*

University of Maryland, College Park — Aug 2022 – May 2025
B.S. Computer Science & Mathematics, Minor Computational Finance — *GPA: 3.9/4.0*

Experience

Yale University — New Haven, Connecticut
Pre-Doctoral Research Fellow, School of Management — *Aug 2025 – Present*
with Bryan Kelly, Head of Machine Learning at AQR

- Investigating and benchmarking whether frontier large language models replicate expert-level investor judgment by having them read Dow Jones Newswires articles and classify the relevant global futures markets.
- Developing a dynamic macroeconomic knowledge graph built from financial news and researching graph algorithms to investigate Shiller's famous "excess volatility" puzzle, linking the evolving flow of macro information to asset prices.
- Building and publishing "Global Macroeconomic Data" (the macroeconomic analogue to Jensen, Kelly, and Pedersen 2023), covering returns of 92 futures contracts using data from LSEG Tick History, LSEG Datastream, and Compustat.

Independent Trading – Kalshi — New Haven, Connecticut
Proprietary Prediction Markets Trader — *Jan 2026 – Present*

- Generated a 19% year-to-date net return on $14,000 of personal capital with a Sharpe ratio of 2.2.
- Deploying an automated market making system in Rust trading across WTI crude oil, Bitcoin, and other commodities markets with real-time position tracking and risk controls.
- Using live CME Globex MDP 3.0 options data from Databento to implement pricing, volatility modeling, and delta hedging.
- Building a sportsbook that quotes same-game parlays on Kalshi, pricing correlated legs off FanDuel's parlay odds and quoting lines into live request-for-quote auctions to capture the vig across NBA, WNBA, MLB, and World Cup matches.

Amazon — Seattle, Washington
Software Engineering Intern — *Jun 2025 – Aug 2025*

- Spearheaded a real-time performance analytics project in Java, presenting key metrics to over 2 million active Amazon Sellers; led a database migration from Elasticsearch to AWS Timestream, projected to cut costs by 10% and user latency by 5%.
- Received a full-time return offer for the Amazon Video Shopping Experiences team.

University of Maryland, College Park — College Park, Maryland
Research Assistant, Department of Finance — *Jan 2024 – May 2025*
with Serhiy Kozak

- Designed a class of latent, three-dimensional factor models ("Tensor Factor Models") motivated by tensor decompositions, capturing drivers of asset returns across time, lags, and cross-sectional dimensions; implemented the model in Python using a dataset of 96 monthly factor portfolios.
- Implemented GPU-accelerated Orthogonalized Alternating Least Squares and PARAFAC algorithms in Jax, enabling scalable factorization of high-dimensional firm characteristics data.

Capital One — McLean, Virginia
Software Engineering Intern — *Jun 2024 – Aug 2024*

- Built a Slackbot using FastAPI for the Data Remediation team, supporting Capital One's internal workflows and enabling faster resolution tracking on data lake platforms; improved query speeds by 15% and cut costs by 5%.

Northrop Grumman — Boulder, Colorado
Software Engineering Intern — *Jun 2023 – Aug 2023*

- Developed a time-varying atmospheric attenuation model with Chebyshev polynomials & approximation theory in PyTorch.
- Created a platform to generate atmospheric attenuation reference tables using Pandas and the Modtran C++ API.

Profile

Technical Skills: Python, Rust, Java, C++, CUDA, Polars, Jax, PyTorch, Huggingface, vLLM, NumPy, Pandas, LaTeX
Teaching Experience: Teaching Assistant for BUFN402 (Portfolio Management)
Professional Development: Yale School of Management Weekly Seminars in Finance, Yale Tobin Center Weekly Pre-Doctoral Seminars in Economics, Yale Empirical Asset Pricing Reading Group
Research Interests: Financial Machine Learning, Global Macro, Commodity Markets, Prediction Markets, Large Language Models, Market Microstructure, Random Matrix Theory